

Feasibility of Quantum Synthetic Matter Platforms as Ontological Probes for Entanglement-First vs. Field-First Descriptions

Input Question

Is quantum many-body entanglement more fundamental than quantum fields?

Domain

Physics

Validation Status

needs_data

Problem Statement

The question asks whether quantum many-body entanglement is ontologically more fundamental than quantum fields. Current literature discusses force unification at high energies and the role of entanglement in phenomena like superconductivity, but lacks direct evidence comparing the fundamentality of these two concepts. The provided evidence highlights Quantum Synthetic Matter (QSM) platforms as precise tools for simulating many-body systems, suggesting a pathway to test if gauge structures (fields) emerge from constrained entanglement patterns.

Rationale

Determining the ontological hierarchy between entanglement and fields is crucial for quantum gravity and high-energy physics foundations. While direct proof is absent, QSM platforms offer a feasible testbed to investigate if field-like symmetries emerge naturally from entanglement-constrained states (entanglement-first) or require explicit field terms (field-first). This study reframes the philosophical question into a falsifiable feasibility study using tensor network simulations validated by exact diagonalization.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Quantum Synthetic Matter (QSM) platforms can serve as feasible testbeds for distinguishing entanglement-first vs. field-first ontologies only if specific tensor network ansätze with intrinsic gauge constraints reproduce effective field theory observables without explicit geometric priors.
- **Mechanism**: If entanglement structures are ontologically prior to fields, then physically valid tensor network states (e.g., Gauge-Invariant PEPS) constructed solely from local entanglement constraints should naturally exhibit emergent gauge symmetries and area-law entanglement scaling characteristic of lattice gauge theories. Conversely, if such features only emerge when explicit field-theoretic terms are added to the

Hamiltonian, this suggests fields are fundamental and entanglement is derivative. This reframes the question as a feasibility study of QSM as an ontological probe rather than a direct proof of fundamentality.

- ****Falsifiable Prediction****: In exact diagonalization or small-scale tensor network simulations, a gauge-invariant PEPS ansatz optimized purely under local symmetry constraints (without geometric input) will fail to reproduce standard lattice gauge theory correlation functions or entanglement entropy scaling unless explicit field-like terms are included in the variational optimization. If it succeeds without such terms, the hypothesis gains support; if it fails, the 'entanglement-first' feasibility is weakened.
- ****Required Observations****: Exact diagonalization validation on small lattices ($N < 20$) comparing gauge-invariant PEPS vs. standard Kogut-Susskind Hamiltonians ; Measurement of entanglement entropy scaling and Wilson loop expectation values in both models ; Quantification of symmetry fidelity under local non-symmetric perturbations to distinguish robust emergence from fine-tuning
- ****Risk of Being Wrong****: Current evidence (EV-Q082-436e9238afd30b4e718a6065) only validates QSM platform precision, not the theoretical link between entanglement patterns and emergent field ontology; observed correspondences may be simulation artifacts or limited to specific ansatz choices rather than universal principles.

Hypothesis 2

- ****Hypothesis****: Categorical quantum mechanics frameworks based on quantum sets and nonlocal games cannot operationally distinguish entanglement-first from field-first ontologies without additional physical axioms beyond synchronicity conditions.
- ****Mechanism****: While categorical constructions using symmetric dagger Frobenius algebras (EV-Q082-3289bc5ce5e77f586863a5f3) provide formal tools for quantum foundations, they lack inherent spacetime or field structure. If entanglement were fundamentally prior, these frameworks should derive relativistic causality or gauge invariance from pure information-theoretic constraints. Failure to do so without ad hoc field-theoretic inputs would indicate that categorical approaches are mathematically equivalent but ontologically neutral.
- ****Falsifiable Prediction****: No derivation of Lorentz covariance or local gauge symmetry exists within the pure quantum game synchronicity framework of Goldberg et al. (EV-Q082-3289bc5ce5e77f586863a5f3) without introducing external field-theoretic axioms. If such a derivation is demonstrated, the hypothesis is falsified; if not, it supports the view that categorical methods alone cannot resolve the ontological priority question.
- ****Required Observations****: Formal analysis of whether quantum set constructions in EV-Q082-3289bc5ce5e77f586863a5f3 yield relativistic or gauge-invariant observables ; Verification that no implicit background geometry or field variables are assumed in the categorical framework ; Comparison of derived correlation structures against standard QFT predictions
- ****Risk of Being Wrong****: The absence of current derivations may reflect incomplete exploration rather than impossibility; future extensions of categorical QM might bridge this gap without new physical axioms.

Technical Details

This research plan is a feasibility study for using Quantum Synthetic Matter (QSM) platforms as ontological probes, explicitly acknowledging that current evidence (EV-Q082-436e9238afd30b4e718a6065) only validates platform control precision, not the theoretical link between entanglement and field ontology. The core technical approach replaces the ill-defined 'entanglement-maximizing' baseline with physically rigorous Gauge-Invariant Projected Entangled Pair States (GI-PEPS). We will compare two distinct ansätze: (1) A GI-PEPS model where

gauge symmetry is enforced strictly through local tensor constraints (representing an 'entanglement-structure-first' construction), and (2) A standard Kogut-Susskind Lattice Gauge Theory (LGT) Hamiltonian simulation (representing a 'field-first' construction). The study focuses on small-scale exact diagonalization ($N < 20$) to validate that chosen entanglement metrics (e.g., topological entanglement entropy, Wilson loop expectations) can discriminate between these constructions before scaling to larger QSM simulations. Claims regarding 'emergent spacetime' or 'ontological priority' are treated as unverified interpretive hypotheses, not established facts.

Datasets

Source

```
```json
{
 "name": "QSM Platform Capability Reference & Synthetic Tensor Network Data",
 "description": "Reference data confirming QSM control precision (EV-Q082-436e9238afd30b4e718a6065) combined with synthetic data generated from exact diagonalization and GI-PEPS simulations of small lattice systems ($N < 20$).",
 "access_note": "QSM capability referenced in EV-Q082-436e9238afd30b4e718a6065. Simulation data will be generated locally using open-source tensor network libraries.",
 "evidence_ids": [
 "EV-Q082-436e9238afd30b4e718a6065"
]
}
```

### Target

```
```json
{
  "name": "Discriminative Observables for Ontological Probes",
  "description": "Calculated metrics including Topological Entanglement Entropy (TEE), Wilson Loop expectation values, and symmetry fidelity scores under local perturbations. These metrics serve as proxies to test if gauge structures emerge naturally from constrained entanglement ansätze.",
  "format": "JSON/CSV containing system size, bond dimension, TEE values, Wilson loop averages, and perturbation response curves."
}
```

Paper Abstract

Background: The ontological relationship between quantum many-body entanglement and quantum fields remains unresolved, with current evidence limited to phenomenological applications. Methods: We propose a feasibility study using Quantum Synthetic Matter (QSM) platforms, leveraging their precision in controlling many-body systems. We implement Gauge-Invariant Projected Entangled Pair States (GI-PEPS) to test if gauge symmetries emerge from local entanglement constraints without explicit field terms. Validation Plan: Small-scale exact diagonalization ($N < 20$) will validate tensor network results, measuring Topological Entanglement Entropy and Wilson Loop expectations. Results: Pending execution of verification experiments. This study aims to determine if QSM can operationally distinguish between entanglement-first and field-first ontologies.

Methods

1. Theoretical Validation: Perform exact diagonalization on small lattices ($N < 20$) for GI-PEPS and Kogut-Susskind models. 2. Model Construction: Implement GI-PEPS tensors enforcing local gauge invariance via symmetry constraints. 3. Simulation: Use DMRG/TEBD for 1D/2D systems to compute ground states. 4. Metric Calculation: Compute Topological Entanglement Entropy and Wilson Loop expectation values. 5. Robustness Test: Apply local non-gauge-invariant perturbations to measure decay of topological order parameters.

Experiments

Baselines

```
```json
[
 "Standard Kogut-Susskind Lattice Gauge Theory (LGT): Explicitly includes field variables and gauge constraints in the Hamiltonian.",
 "Random Tensor Network (RTN) Control: A model with no physical constraints, to establish baseline entanglement scaling unrelated to gauge theory."
]
```

### Metrics

```
```json
[
  "Topological Entanglement Entropy (TEE): Quantifies long-range entanglement characteristic of topological order.",
  "Wilson Loop Expectation Value: Measures gauge-invariant correlation functions.",
  "Symmetry Fidelity Score: Overlap of the perturbed state with the gauge-invariant subspace."
]
```

'''

Ablation

Vary the bond dimension of the GI-PEPS ansatz and the strength of explicit symmetry-breaking perturbations to determine the threshold at which emergent gauge features collapse.

Validation Protocol

Cross-validate all tensor network results with exact diagonalization for $N < 20$. Ensure that observed topological signatures are not finite-size artifacts by checking convergence with increasing system size L .

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- **EV-Q082-436e9238afd30b4e718a6065** · arxiv · arXiv:2112.04501
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2112.04501.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=77850425fa26cec9df9b9aaa23278995b2cfd0360e24a6a670bd5e6e825b69cc

Reviewer Comments

- The revised hypothesis correctly reframes the inquiry as a 'feasibility study' for QSM-based ontological probes, directly addressing the critical issue regarding the lack of evidence for a causal link between QSM and fundamental ontology (EV-Q082-436e9238afd30b4e718a6065).
- The replacement of the ill-defined 'Entanglement-Driven' baseline with Gauge-Invariant PEPS (GI-PEPS) resolves the strawman comparison concern; this is a physically rigorous ansatz grounded in established tensor network literature for lattice gauge theories.
- The inclusion of mandatory exact diagonalization validation ($N < 20$) and specific discriminative metrics (TEE, Wilson Loops) satisfies the requirement for theoretical validation before scaling, ensuring reproducibility and falsifiability.
- All claims regarding emergent spacetime or ontological priority are now explicitly labeled as unverified interpretive hypotheses, eliminating overclaiming risks identified in the previous review.
- Results are correctly marked as pending, and no fabricated data or unauthorized references are present.

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Provide source code for GI-PEPS construction and exact diagonalization scripts.
- Specify exact lattice geometry, group structure (e.g., Z_2 , $U(1)$), and bond dimensions.
- Document the method for calculating Topological Entanglement Entropy (e.g., Kitaev-Preskill or Levin-Wen prescription).
- Include raw data for Wilson loop expectations and symmetry fidelity scores.
- Detail the perturbation protocol used for robustness testing.